EXCEL IN O' LEVEL

CHEMISTRY

A simplified, well elaborated and illustrative approach to the chemistry Competence Based Curriculum

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School	
Class	

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Table of Contents

CHEMISTRY AND SOCIETY

Introduction to chemistry

Chemistry is a branch of science that deals with the composition, structure, properties and behavior of matter.

Chemistry finds out what substances contain, how they react with each other, how they vary under different conditions, and the energy evolved or taken in during those changes.

Chemistry is about atoms and molecules which are the building blocks of everything around us. This explains why soap cleans, iron rusts, what happens when food is cooked and how medicine works.

It focuses mostly on substances, what they consist of and how they do change.

It is a discrete a science for it owns specific principles, wording like chemical equations and unique methods from other branches of science.

It includes man's attempt to transform the natural world in order to benefit from nature's hidden resources.

Common chemicals in the environment

Most of the products used by man in everyday life are combined outcomes of the knowledge of chemistry in the various fields like pharmaceuticals, cosmetics, plastics, food and beverages, soaps and detergents, water treatment and many others.

Below are suggested common chemicals in man's everyday life.

FIELDS	PRODUCTS	SOURCE
Pharmaceuticals and cosmetics	Paracetamol or panadol	Crude oil, chemical synthesis.
	Aspirin	Petroleum, sulphuric acid.
	Perfumes	Ethanol, vanilla, citrus peels, animal chemicals like
	Skin creams	Petroleum, cocoa beans, emulsifiers, preservatives.
	Lipstick	Bee wax, castor beans, iron oxides, petroleum.
Plastics	Water bottles	Crude oil, ethene.
	Polythene bags	Crude oil, ethene
	Pipes (PVC)	Crude oil, common salt, stabilizers, plasticizers.
	Cups (Styrofoam cups)	Crude oil, styrene monomer (from benzene and ethylene)
Food and beverages	Table salt (common salt)	Brine, iodine (from iodized salt), magnesium carbonate.

EXPERIMENTAL CHEMISTRY

A laboratory

This is a special room where scientific experiments are conducted from and apparatus are kept.

Chemistry laboratory rules and regulations

Students should not enter the laboratory before the teacher tells them to do so.

Laboratory materials should be used on purposes directed by the teacher only.

Students should never taste anything in the laboratory.

Bottles should never be handled by their necks.

Running is highly prohibited in the laboratory.

Always wash or clean the apparatus and your hands after every chemical experiment.

Report any accident to the teacher/technician immediately.

Do not carry out any experiments that are not authorized by the teacher.

Place broken glasses and solids into trash bins situated in the laboratory.

When your experiment is completed, turn off the water supply and gas supply and disconnect any electrical connections and return all the materials and apparatus in their proper places.

Reasons for observing the chemistry laboratory rules

To prevent accidents and injuries.

To avoid chemical contamination.

To protect laboratory equipment.

To ensure accurate experimental results.

To promote discipline and responsibility.

To prevent fires and explosions.

How best should you deal with laboratory fires?

Laboratory fires occur mainly due to unsafe handling of flammable substances and poor safety practices. Never the less, chemistry has in place the fire safety guidelines as illustrated below;

Do not run in the laboratory.

Keep flammable substances away from flames.

Turn off the Bunsen burner/fire source when not in use.

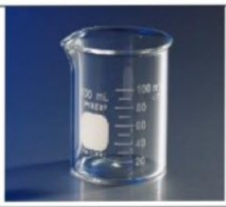
Be aware of the escape routes in case of a fire outbreak.

Report chemical spills immediately.

Use fire extinguishers in case of small fires.

Steps taken in using a fire extinguisher (PASS METHOD)

P- Pull the safety pin.

Reagent bottles		To hold liquid chemicals.
Retort stand		To support apparatuses like burettes.
Round bottomed flasks		A glass flask used in a laboratory for holding chemicals which have a spherical shape for uniform heating.
Separating funnel		Used to separate immiscible liquids which do not mix, like oil and water.
Mortar and pestle		For pounding and grinding solid substances.
Beaker		To mix, hold, and pour liquid or powdered substances.
Boiling tube		To strongly heat substances on the flame of a heat source.

Conclusion

Water separates from a salt solution due to its very low boiling point (100^0c) as compared to that of salt (1400^0c).

Separation by fractional distillation

This is used when separating two components of miscible liquids with a small difference in their boiling points for example ethanol (78^0c) and water (100^0c).

Other components that can be separated by this method include; petrol, diesel, kerosene; oxygen and nitrogen etc.

Aim: An experiment to investigate the separation of ethanol from water by fractional distillation.

Hypothesis: Heating a mixture of two miscible liquids makes the liquid with a lower boiling point distill first for it evaporates faster than the one with a higher boiling point.

Independent variable: The number of fractions/portions collected.

Dependent variable: The concentration of ethanol in each fraction/boiling temperature during distillation.

Controlled variable: The initial volume and composition of the mixture.

Apparatus and materials: Distillation flask, fractionating column, thermometer, condenser, retort stand, heat source, receiving flask.

Procedures: Poured the mixture in the flask; properly set up the distillation apparatus with a fractionating column between the flask and condenser; started the flowing of the cold water in the condenser; gently heated the solution; collected the first fraction(ethanol) at its boiling point; increased temperature to collect the second fraction(water).

Risk(s) and mitigation(s)

Burns from the hot apparatus; use heatproof gloves.

Flammable liquids like ethanol; use water bath for flammable liquids.

Fire hazard from open flames; keep away the flammable materials.

Results obtained

First liquid collected at lower temperature; temperature rises before second liquid is collected.

Results' interpretation

Liquids have different boiling points; fractionating column improves separation by vaporization and condensation.

Criteria for purity of substances

The purity of a substance can be determined by measuring the;

Boiling point e.g., 100^0 c for water.

Melting point e.g., 0^0 c for water.

Density e.g., 1 g/cm^3 for water.

Refractive index e.g., 1.33 for water.

Determination of boiling and melting points.

These are physical properties of substances. They help us identify substances and determine their purity.

Melting point is the temperature at which a solid turn into a liquid.

This occurs at a fixed temperature for a pure substance. During melting, temperature remains constant until all solid melts.

Impurities usually lower the melting point and make melting occur over a range.

Ice melts at 0^0c under normal atmospheric pressure.

The melting point of substances can be experimentally determined as described below;

Aim: An experiment to determine the melting point of naphthalene by heating it gradually and observing the temperature range over which it changes from solid to liquid.

Hypothesis: If pure Naphthalene is heated, then it will melt at a fixed temperature because pure substances have sharp melting points.

Independent variable: The time of heating.

Dependent variable: The temperature of naphthalene.

Controlled variable: Amount of the naphthalene.

Apparatus and materials: Boiling tube, water, glass beaker, tripod stand, naphthalene, heat source, stop watch, thermometer.

Procedures: A test tube was filled with 4g of naphthalene and a thermometer inserted; the test tube was suspended in a glass beaker half filled with water; the level of naphthalene was kept below that of water in the beaker; water was heated and slowly stirred naphthalene with a thermometer; the stop watch was started immediately when naphthalene started melting and the temperature changes recorded every 30 seconds until all the substance melted; the temperatures were recorded against the times at which they occurred and a graph of temperature against time plotted.

Risk(s) and mitigation(s)

Burns from hot flames; use water bath to avoid uneven heating.

Cuts from broken glassware which may lead to injuries and pain; proper use of protective gears.

Results obtained

Temperature(^{0}c)	60.0	67.0	74.0	79.0	80.0	81.0	81.5	82.0	85.0
Time(s)	0.0	30.0	60.0	90.0	120.0	150.0	180.0	210.0	240.0

SALTS

A salt is the substance formed when either all or part of the ionisable hydrogen ions of an acid are replaced by a metallic ion or ammonium ion.

A salt is formed when an acid reacts with base.

Salts exist naturally in the earths crust with lots of impurities and therefore costly to purify using chemical means.

Examples of salts are summarized in the table below.

The table shows the salt and the acid from which it is prepared from.

Salt	Formula	Acid
Sodium chloride	$NaCl$	Hydrochloric acid
Potassium chloride	KCl	Hydrochloric acid
Ammonium chloride	NH_4Cl	Hydrochloric acid
Silver nitrate	$AgNO_3$	Nitric acid
Copper (II) sulphate	$CuSO_4$	Sulphuric acid
Magnesium sulphate	$MgSO_4$	Sulphuric acid
Sodium hydrogen sulphate	$NaHSO_4$	Sulphuric acid
Sodium sulphate	Na_2SO_4	Sulphuric acid
Sodium carbonate	Na_2CO_3	Carbonic acid
Calcium carbonate	$CaCO_3$	Carbonic acid
Sodium hydrogen carbonate	$NaHCO_3$	Carbonic acid

Types of salts

Salts can be divided into two main groups i.e., normal salts and acid salts.

Normal salts are salts formed when all the replaceable hydrogen in an acid has been replaced by a cation e.g., sodium chloride, sodium carbonate.

Acid salts are salts formed when part of the replaceable hydrogen in an acid has been replaced by a cation e.g., potassium hydrogen sulphate, sodium hydrogen sulphate, sodium hydrogen carbonate.

Properties of salts

1. Physical state at room temperature. Salts are mainly solids at room temperature though some are hydrated and thus crystalline. Others are anhydrous and appears as powders for instance the carbonates.

Most salts are white whereas some have distinct colors. Their colors indicate the cations present in the salt.

2. Electrical conductivity. All salts being ionic compounds conduct electricity when in aqueous solution or molten state.

REACTIVITY SERIES

Reactivity series refers to the arrangement of elements from the most reactive element to the least reactive element.

The series is based on the reactions with air, water and dilute acids.

Reactivity is greatly influenced by atomic size.

Reaction of metals with water.

Aim: An experiment to investigate the reaction of potassium and sodium metals with water.

Materials and apparatus: Litmus paper, potassium metal, sodium metal, water, filter papers, trough, knife, pair of tongs.

Procedure: Water was poured in the trough; a tiny piece of sodium was cut and cleaned or made oil free using a filter paper; dropped it with a pair of tongs into a trough of cold water; carefully observed and noted; tested the liquid in the trough with red and blue litmus papers and noted the colour changes; repeated the above steps using potassium.

Observation(s)

Sodium melts into a silvery ball and darts on the surface of water with a hissing sound giving off hydrogen gas and sodium hydroxide. This turns red litmus paper blue.

Potassium reacts vigorously with cold water; melts and darts on the surface of water with a hissing sound giving off hydrogen gas to form potassium hydroxide solution which turns red litmus blue.

Data Analysis

The hissing sound is due to evolution of hydrogen gas. Sodium hydroxide and hydrogen gas are formed.

Word equation

Sodium + Water →Sodium Hydroxide + Hydrogen

Chemical equation

$2Na\ (s) + 2H_2O\ (l) \rightarrow 2NaOH\ (aq) + H_2\ (g)$

Word equation

Potassium + Water → Potassium Hydroxide + Hydrogen

Chemical equation

$2K\ (s) + 2H_2O\ (l) \rightarrow 2KOH\ (aq) + H_2\ (g)$

Reaction of other metals with water.

Materials and Apparatus: Beaker, filter funnel, boiling tube, water, calcium metal, magnesium, aluminium, zinc, iron, copper, spatula.

Procedure: Put a spatula endful of calcium at the bottom of the beaker containing some water; inverted a funnel over calcium metal; added enough water to cover the whole funnel; quickly inverted a boiling tube full of water and collected the gas; carefully observed what happened to the metal; the boiling tube was corked under water when it was full of hydrogen

Alloys	Properties and uses	Environmental impacts and mitigation
Bronze (copper and zinc)	It is very hard; for making coins and medals. It is strong and tough/ has high tensile strength; for manufacture of marine hardware like propellers, ship fittings. It has a very high density. It is corrosion resistant. It is a good conductor of heat and electricity. It is malleable and ductile. It has a very high melting point.	Accumulation of bronze waste in the soil can lead to absorption by plants that are consumed by people. This may lead to cancer hence death; **mitigation** is by recycling of bronze waste. It may cause cancer when in contact with drinking water and therefore death; **mitigation** is by recycling of bronze waste. Its waste is non-biodegradable hence causes soil pollution. This leads to low crop yields; **mitigation** is by recycling of bronze waste.
Solder (Tin and lead)	It is soft, malleable, strong. It is a good conductor of heat and electricity; used in making electrical appliances. It has a high density. It has a low melting point; making of plumbing joints.	Its waste is non-biodegradable hence causes soil pollution. This leads to low crop yields; **mitigation** is by recycling of solder waste.
Brass (Copper and zinc)	It is shiny/lustrous; used to make decorative items, door handles. It is strong and tough. It is corrosion resistant; used to make plumbing fittings, taps. It is malleable and ductile; for manufacture of musical instruments like trumpets. It is a good conductor of heat and electricity; used to make electrical connectors and terminals. It has a very high density. It has a very high melting point.	It may cause cancer when in contact with drinking water and therefore death; **mitigation** is by recycling of brass waste.
Duralumin (Aluminium, copper and magnesium)	It has a low density or it is light in weight; used to make aircraft and aerospace parts. It is a good conductor of heat and electricity; used to manufacture kitchen utensils.	Accumulation of bronze waste in the soil can lead to absorption by plants that are consumed by people. This may lead to cancer hence death; **mitigation** is by recycling of duralumin waste.

STATES AND CHANGES OF STATES OF MATTER

Matter

Matter is anything that occupies space and has weight.

States of matter

The states of matter are solids, liquids, gases and plasma.

Solids

The particles of solids are closely and regularly packed together.

They are held together by strong forces which attract particles to each other.

At a certain temperature, the regular arrangement of particles in the solid state breaks down.

At this point the solid melts and turns into liquid state. The temperature at which solids change to liquid is called **melting point.**

The characteristics of solids include: fixed volume, fixed shape, incompressible and cannot be poured.

Lliquids

The particles are close together, the attraction forces are moderate, particles slide past each other and there is random particle movement.

The properties of liquids include; a fixed volume, have no definite shape (takes the shape of a container), can be poured and are not easily compressible.

When liquids are heated their particles move faster and finally turn into gas.

The temperature at which a liquid change into a gas is called the **boiling point** e.g., the boiling point for pure water is 100^0c

Gases

The particles are very far apart due to weak forces of attraction, they freely move at a high speed and randomly which makes them collide with each other and with walls of the container.

The characteristics of gases include; do not have a definite shape and volume; highly compressible; very low density; can flow easily and the particles diffuse at a very high rate.

When energy is supplied to gases through very high temperatures/powerful radiation or strong electric current, gas particles turn to the plasma state of matter.

Examples of gases include;

Hydrogen, helium, nitrogen, oxygen, fluorine, neon, chlorine, and argon.

Plasma

This is a hot ionized gas made up of positive ions and free electrons.

USING MATERIALS

A material refers to a substance with specific characteristics from which objects, tools, fabrics, and buildings are made.

Categories of materials.

There are two classes of materials namely; natural and synthetic/artificial materials.

Natural materials are materials made by God/exist in nature.

Examples include; cotton, wool, silk, wood, stones and many others.

Properties of natural materials.

These show various properties that affect and dictate their usefulness in the environment. These include characteristics like biodegradability, light weight, flexibility, hardness, strength, thermal and electrical insulation, durability and many others as elaborated in the table below.

Materials	Category	Properties	Uses
Wood	Natural	Readily available thus easy to acquire cheaply. Strong thus can carry heavy loads. Light when dry therefore good for roofing. Easy to smoothen to give better appearance. Can be destroyed by termites or rot when untreated.	Building, furniture, fuel, tools.
Paper	Natural	Can easily be decomposed by bacteria/ it is biodegradable. It is foldable. Absorbs moisture. Stiff, strong.	Making packaging materials for dry products.
Stones	Natural	It is very durable and strong, hard and heavy, resistant to weathering. Has high compressive strength.	For building of foundations, bridges, roads, floors and monuments. For decorations
Sand	Natural	It has fine granular particles. Doesn't shrink or expand easily. It gives strength when mixed.	For wall plastering, concrete production, glass manufacture, filling and leveling the ground, making mortar paste i.e., a mixture of sand, cement and water.

Classification of substances.

A substance refers to a form of matter with a fixed composition and distinct properties.

Substances can be classified as elements, compounds and mixtures.

Elements

An element is a pure substance that cannot be split into simpler substances by chemical means.

Examples include all elements in the periodic table like hydrogen, boron, oxygen, neon, sodium, silicon, chlorine, potassium.

Below is a list of the first 20 elements in the periodic table in order of their atomic numbers and symbols.

Students are required to memorize the elements below for better understanding of the forth coming chapters in the whole course.

Atomic number	Song (that helps memorize the elements in their order)	Element	Symbol
1	He	Hydrogen	H
2	Henry	Helium	He
3	Loves	Lithium	Li
4	Betty	Beryllium	Be
5	But	Boron	B
6	Can	Carbon	C
7	Not	Nitrogen	N
8	Offer	Oxygen	O
9	Full	Fluorine	F
10	Needs	Neon	Ne
11	So	Sodium/Natrium	Na
12	Mary	Magnesium	Mg
13	And	Aluminium	Al
14	Sam	Silicone	Si
15	Played	Phosphorus	P
16	Soccer	Sulphur	S
17	Carelessly	Chlorine	Cl
18	And	Argon	Ar
19	People	Potassium/Kalium	K
20	Came	Calcium	Ca

CHEMICAL REACTIONS

Rate of reaction.

Some reactions proceed very fast. For example, the addition of aqueous ammonia to a solution of lead (II) salts forms a white precipitate immediately.

Some reactions proceed at a moderate speed. It takes sometime for the reaction between calcium carbonate and dilute hydrochloric acid to go to completion.

Other reactions are slow. For instance, it takes iron a few days to rust in moist air.

The above-mentioned reactions occur at distinct speeds.

The rate of a chemical reaction is the progress of the reaction in unit time.

Alternatively, the rate of a chemical reaction is the rate at which products are formed or the rate at which reactants are used up in the reaction.

$$\text{Rate of reaction} = \frac{\text{concentration in moles per litre}}{\text{Time in seconds}}$$

Units for the rate of reaction are moles/litre/second (mol/l/s)

Factors affecting the rate of chemical reactions.

Factors that affect the rate of a chemical reaction include; concentration, temperature, surface are (particle size), pressure, catalyst and light.

<u>Effect of concentration of reactants on the rate of reaction.</u>

The rate of the reaction depends on the number of times that the reacting particles collide. This depends on the concentration of the reactants.

The higher the concentration, the higher the number of collisions and therefore the higher the rate of chemical reaction.

<u>Investigating the effect of temperature on the rate of reaction.</u>

Aim: An experiment to investigate the effect of concentration on the rate of reaction between sodium thiosulphate and hydrochloric acid.

Hypothesis: Decrease in concentration of sodium thiosulphate slows down the rate of reaction.

Independent variable: The concentration of sodium thiosulphate.

Dependent variable: Time taken for the formation of Sulphur or time taken for the mark, X to disappear.

Controlled variable: The volume and concentration of hydrochloric acid.

Materials and apparatus: Conical flask/beaker, white tile/paper, stop clock/watch.

Procedure: Made a mark, X with blue or black ink on a piece of paper; placed 50 cm^3 of 1.5 M sodium thiosulphate solution into a beaker; added 10 cm^3 of 0.1 M hydrochloric acid to the

sodium thiosulphate and started the stop clock immediately; gently shaked the mixture to mix the solution well and placed the beaker above the mark on the paper; watched the mark

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Task;

(a) Design an experiment that can be used to determine the amount of heat produced.

(b) Carry out the experiment and determine the amount of heat liberated.

(c) Analyse the results and draw your conclusion(s).

Worked out item

An organization dealing in fishing around river Nile prepared a workshop to train local fish dealers on making table salt on a small scale for fish preservation. This involved mixing sodium hydroxide and hydrochloric acid. During this, a member was randomly appointed to add a prepared solution of acid to a base solution in a container and noted some change in temperature but kept on adding the acid. He had no clue on what was happening.

Sodium hydroxide and hydrochloric acid react according to the following equation.

$NaOH\ (aq) + HCl\ (aq) \rightarrow NaCl\ (aq) + H2O\ (l) + Heat$

The heat produced changes with volume of the acid added to the base.

You are provided with FA_1 or acid and FA_2 or base.

Task;

As a knowledgeable chemistry learner;

a) **Design an experiment you will carry out to determine the heat amount produced.**

Aim: An experiment to determine the maximum amount of heat produced during the reaction between FA_1 and FA_2.

Hypothesis: The reaction between FA_1 and FA_2 is exothermic or liberates heat.

Dependent variable: Temperature of the solution.

Independent variable: Volume of the acid added.

Controlled variable: Volume of FA_2 measured.

Risks	**Mitigations**
Use of damaged apparatus that may cause cuts.	Proper inspection of equipment before use.
Irritation due to pouring of the acid solution on the skin.	Wearing a lab coat, gloves and closed shoes.
Breakage of the thermometer.	Putting back the thermometer in its case after use.

Procedure: Pipetted 25cm^3 of FA_2 into a plastic beaker, noted and recorded its initial temperature; also noted and recorded the temperature of FA_1, filled it into the burette and adjusted to the zero mark; FA_1 is run to FA_2 in the beaker at uniform intervals of 5cm^3 each time stirring and noting the highest temperature of the mixture for at least seven readings.

b) **Carry out the experiment and record your findings.**

Initial temperature of FA_1=

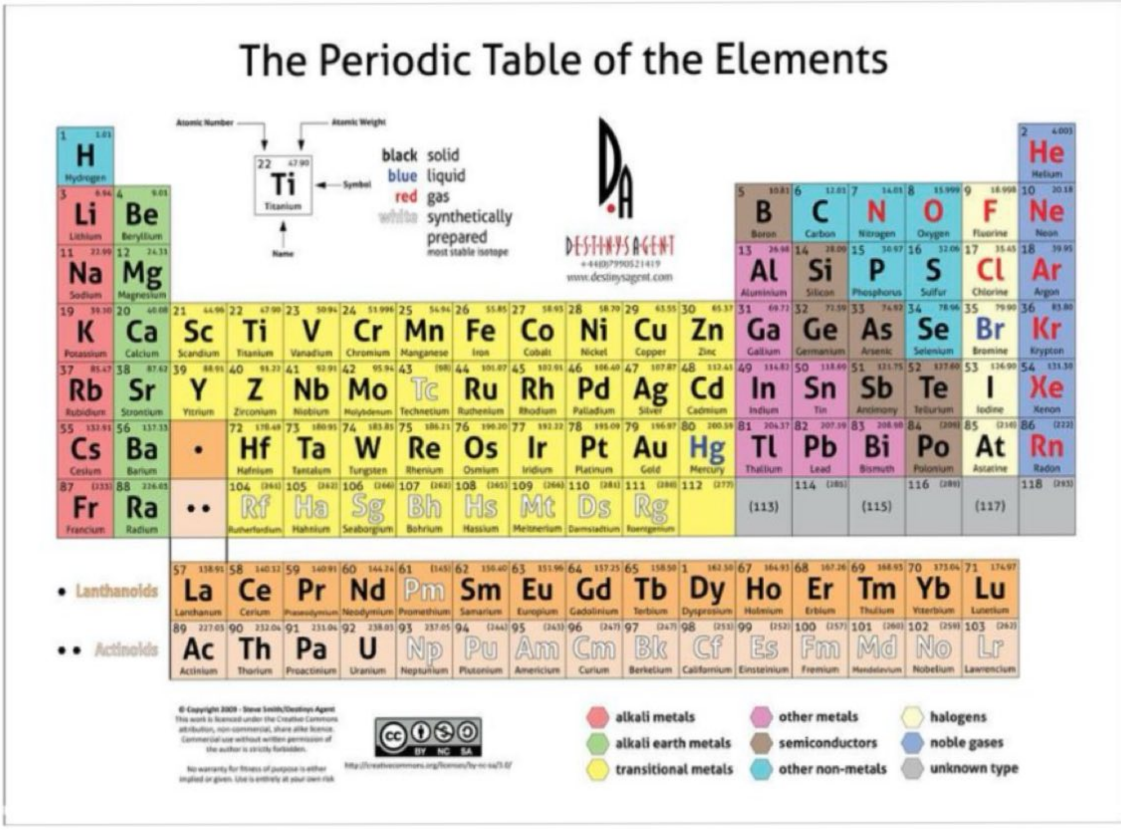
Initial temperature of FA_2=

Average initial temperature of FA_1=

Volume of FA_2 used = 25.0 cm^3

Volume of FA_1 added/cm^3	0	5	15	20	25	30	35
Highest	Room						

THE PERIODIC TABLE



The periodic table refers to the tabular arrangement of chemical elements, organized by increasing atomic number, electron configuration, and recurring chemical properties.

It is arranged into groups and periods.

Periods are the horizontal rows within the periodic table.

Elements in the same period have the same number of electron shells.

Groups are the vertical columns within the periodic table.

Elements within the same group share similar valence electron configurations and thus exhibit analogous chemical behavior.

The periodic table highlights periodic trends such as electronegativity, atomic radius, ionization energies etc.

Categories of elements in the periodic table.

Elements in the periodic table are categorized into four classes i.e., metals, non-metals, metalloids/semi metals and noble gases.

Metals lose their valence electrons/electrons in their outermost shell to form cations (positively charged ions).

	low temperatures).
Helium has a high thermal conductivity.	Important in leak detection.
Neon has a bright red orange discharge.	Used in neon signs and cryogenic refrigerants.
Argon is inert, abundant and dense.	It is respectively used in welding shield, fluorescent lamps and wine preservation.

Environmental impacts and mitigations of noble gases.

They can displace oxygen in the air leading to suffocation of organisms and hence death. This can be **mitigated by** ensuring proper ventilation in areas where noble gases are stored or used.

Worked out Item.

Learners in S.1 North came across a rare solid substance, M, which they suspected to be an element. 0.3g of the element could burn in air to form 0.5g of the solid product. One of them picked interest in what could be the chemical formula of the oxide of the element. However, he did not know how to determine the formula. When they contacted the laboratory technician, he gave them the atomic number and mass number of M as 12 and 24 respectively, and the symbolic representation of oxygen as ${}^{16}_{8}O$. As a student of chemistry help the learners to;

(a) understand the nature of substance M.

M is a metal, since it forms ions by loss of electrons, for example potassium.

(b) determine the formula of the oxide of M.

Mass of oxygen in the oxide = 0.5 – 0.3 = 0.2g

Symbols of elements present	**X**	**O**
Composition by mass	0.3	0.2
Number of moles	0.3/24	0.2/16
	0.0125	0.0125
Simplest ratio (mole ratio)	0.0125/0.0125=1	0.0125/0.0125 =1

Hence the formula of the oxide is XO.

(b) know the environmental consequences of the element.

When in contact with drinking water, especially in large amounts can cause cancer leading to death, **mitigated by** purifying the water before use.

Sample Item.

A group of learners was faced with unique substances ${}^{23}_{11}W$; ${}^{32}_{16}X$; ${}^{11}_{5}Y$; and ${}^{4}_{2}Z$ which they suspected to be elements. They were concerned about what type of elements they could be, their appropriateness, and environmental impacts. Write a report to address their concerns.

Structure of an atom.

An atom is the smallest electrically neutral indivisible particle of an element which can take part in a chemical reaction.

The fundamental particles of an atom.

An atom is made of three fundamental particles namely; protons, electrons, and neutrons.

Electrons (e). These are negatively charged particles of an atom.

They are arranged around the nucleus of an atom.

Protons (p). These are positively charged particles.

They are found inside of the nucleus of an atom.

Neutrons (n). These are neutral particles and thus have no charge. They are also found inside the nucleus of an atom.

3. Potassium, K

Atomic number = 19

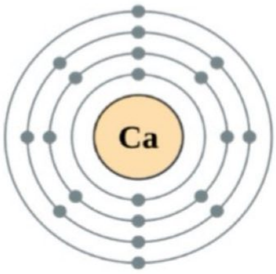
Electronic configuration =

Draw its structure and hence write the configuration.

4. Calcium, Ca

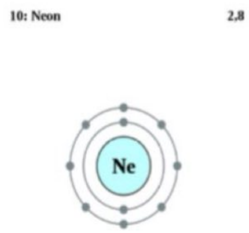
Atomic number =20

Electronic configuration is 2:8:8:2



5. Neon, Ne

Atomic number = 10



Electronic configuration is 2:8

Metals and non-metals.

Metals have 1, 2 or 3 electrons in their outermost shell, non-metals have 4, 5, 6, 7, noble gases have 8 electrons in the outermost shell, and metalloids have 4 electrons.

Stability of elements.

Elements become stable by losing or gaining electrons to have electronic configurations similar to those of noble gases.

Metals become stable by **losing** their outer electrons to remain with 8 electrons in the outermost shell.

c) Water

Response

Elements present	Hydrogen	Oxygen
Symbols of elements	H	O
Interchange of valences	H^1 O^2	
	H_2O_1	
Final formula	H_2O	

Note that the ones (figure 1) don't appear in the formula because they are conventionally taken to be silent.

d) hydrogen peroxide.

Response

Elements present	Hydrogen	Peroxide radical
Symbols of elements	H	O_2
Interchange of valences	H^1 $O_2{}^2$	
	H_2O_2	
Final formula	H_2O_2	

e) Sodium peroxide

Response

Elements present	sodium	peroxide radical
Symbols of elements	Na	O_2
Interchange of valences	Na^1 $O_2{}^2$	
	$Na_2(O_2)_1$	
Final formula	Na_2O_2	

Activity for Trial

Write the formula of the following compounds.

a) Calcium chloride
b) magnesium carbonate
c) lithium sulphate
d) aluminium hydrogencarbonate
e) Potassium hydrogenphosphate
f) calcium dihydrogenphosphate
g) ammonium chloride
h) boron nitrate
i) ammonium carbonate
j) beryllium nitrate

Variation in melting and boiling points of period 3 elements.

Elements of distinct groups and periods portray various trends in melting and boiling points.

Melting and boiling points depend on the strengths of bonds/forces between the particles within a substance.

Element	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Melting point/^{0}c	98	650	660	1414	44	115	-101	-189
Boiling point/ ^{0}c	883	1090	2470	3265	280	445	-34	-186

a) On the same pair of axes, draw a graph of melting and boiling points against atomic number for the elements.

Support material.

Two curves are expected, one for melting point and another for boiling point.

The points should be joined using a ruler.

The points plotted should be labelled with the symbols of elements.

b) State the trends in melting and melting points across the period.
These increase from sodium to aluminium (among metals) and generally reduce from silicon to argon (among non-metallic elements).

c) Which factors contribute to the trends noticed above.
Atomic radius.
Number of valence electrons.

d) Describe the graph and hence explain.
The melting and boiling points increase from sodium to aluminium because of the increase in strengths of their metallic bonds.
Silicon adopts a three-dimensional structure with many interlinking covalent bonds that give it extra stability thus the exceptionally high melting and boiling points.
Phosphorus, Sulphur and chlorine exist as molecules held by van der Waal's forces of attraction, whose strength increase with increase in molecular size.

Variation of density across the period.

Element	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Density (g/cm^3)	0.97	1.74	2.70	2.33	1.82	2.07	0.0032	0.00178

As a knowledgeable chemistry learner,

(a) Help them,

(i) understand the classes of the elements.

X, Y and Z are metals, for they form ions by loss of electrons, for example calcium and potassium.

A, Q and M are nonmetals because they form ions by gain of electrons, for example oxygen, sulphur.

W is a metalloid because it has both metallic and non-metallic properties, for example silicon.

(ii) on how to interpret the data.

There is a sharp increase in melting points from X to Y because of the decrease in atomic radius and increase in the number of electrons each metal atom contributes towards metallic bond formation resulting into stronger metallic bonds.

Melting point increases slightly from Y to Z because of the slight increase in the strength of the metallic bond due to the decrease in the atomic radius.

W has the highest melting point because it adopts a giant atomic structure with many strong covalent bonds that require a lot of heat energy to break.

A, Q and M are non-metals which exist as molecules with simple molecular structures and the melting point decreases with decrease in the strength of the Van der Waals forces due to decreasing molecular mass.

(b) suggest to them the possible uses and environmental impact of the elements

Uses

Metals are used in electronic devices, in construction, manufacture of aeroplanes.

Non-metals are used as electrical insulators, manufacture of drugs, fertilizers.

Metalloids, used in production of alloys, as flame retardants, as semiconductors in the manufacture of dry cells and batteries.

Environmental impacts.

Metals

When in contact with drinking water, especially in large amounts, metals can cause cancer leading to death. This is mitigated by purifying the water.

Their accumulation in soil can be absorbed by plants which are consumed by humans. This can cause cancer leading to death. This can be mitigated by consuming plants with the right amount of mineral content.

When contaminated in air and inhaled, metals may cause respiratory diseases hence death. This is mitigated by purifying the air.

methylene group ($-CH_2$).

Examples of homologous series are alkanes, alkenes, alkynes, alcohols, carboxylic acids etc.

Crude oil and fractional distillation.

Crude oil is a mixture of distinct hydrocarbons found underground.

A hydrocarbon is a compound made of only carbon and hydrogen e.g., alkanes, alkenes and alkynes.

Crude oil mainly consists of alkanes, alkenes, benzene, toluene, traces of metals, Sulphur, oxygen and nitrogen compounds. It has so many impurities and hence purified/refined by fractional distillation.

The impurities include; Sulphur, nitrogen, oxygen compounds and salts.

NB: Purification and uses of crude oil, manufacture of ethanol, natural gas, biogas, polymers, , detergents have respectively been sorted under the topics industrial processes, fossil fuels, using materials and chemicals for consumers.

Alkanes

These are saturated hydrocarbons with a carbon-to-carbon single bond as their functional group.

Alkanes are characterized with a suffix "ane" meaning all alkanes end with that sound i.e., methane, ethane, propane etc.

They have a general molecular formula of: $\mathbf{C_nH_{2n+2}}$ where;

n – is the number of carbon atoms and $n \geq 1$.

Substituting in the formula, $\mathbf{C_nH_{2n+2}}$.

Activity

Generate the molecular formulae of alkanes using the general molecular formula above and fill in the table below.

For n=1; $C_1H_{2(1)+2} \rightarrow C_1H_4 \rightarrow CH_4$

For n=2; $C_2H_{2(2)+2} \rightarrow C_2H_6$

Trials

Try out from n=3 to n=12

Naming alkanes.

This involves adding the suffix "ane" to the parent name of the respective alkane i.e.,

n	Parent name
1	Meth
2	Eth
3	Prop
4	But
5	Pent
6	Hex

Generate the molecular formulae of alcohols using the general molecular formula above and fill in the table below

$\rightarrow CH_3OH$.

For n=7; $C_7H_{2(7)+1}OH \rightarrow C_7H_{15}OH$.

Trials

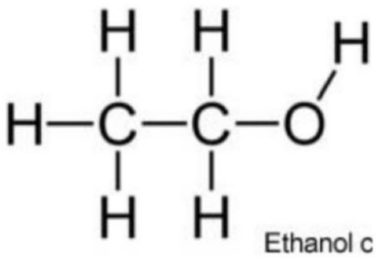
Try out from n=3 to n=13.

Naming alkanols/alcohols.

This involves substituting the letter "e" with the suffix "ol" to the respective alkane i.e., ethane becomes ethanol, octane becomes octanol etc.

The structural formula is written by first displaying all the carbon atoms in that compound, filling in the bonds starting with the attaching of the -OH bond to any carbon atom as the functional group and making sure each carbon atom gets a maximum of four bonds and completing the formula by filling in the hydrogen atoms e.g.

For ethanol, C_2H_5OH



Ethanol c

n	Molecular formulae	Name	Structural formulae
1	CH_3OH	Methanol	
2	C_2H_5OH		CH_3CH_2OH
3			
4		Butanol	
5	$C_5H_{11}OH$		
6			$CH_3CH_2CH_2CH_2CH_2CH_2OH$
7		Heptanol	
8			
9	$C_9H_{19}OH$		
10			
11			$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2OH$
12			
13	$C_{13}H_{27}OH$	Tri-decanol	

Properties of alcohols.

The lower members are too soluble in water but solubility decreases as carbon chain increases.

Their melting points and boiling points are higher than those of the corresponding alkanes and they also increase with increase in molecular size.

The lower members i.e., methanol, ethanol and propanol are liquids and the higher members are waxy solids at room temperature.

Most alcohols are less dense than water and their densities increase with increasing molecular

STRUCTURES AND BONDS

A bond is a force of attraction that holds atoms or ions together to form substances like molecules or compounds.

An element is a substance that cannot be broken down into smaller simpler substances by chemical means for example hydrogen, nitrogen, argon, calcium etc.

A molecule is a group of atoms of the same element or different elements held together by strong covalent bonds when a pair of electrons is shared for example oxygen, chlorine, water, carbon dioxide, nitrogen etc.

Isotopes are atoms of the same element with the same atomic/proton number but distinct atomic mass/mass number.

Isotopy refers to the existence of atoms of the same element with the same atomic number but different atomic mass/nucleon number.

Examples of elements that show isotopy include;

Hydrogen: $^{1}_{1}H$, $^{2}_{1}H$, $^{3}_{1}H$.

All the hydrogen atoms above share the same atomic number but their atomic masses are different.

Carbon: $^{12}_{6}C$, $^{13}_{6}C$, $^{14}_{6}C$.

All the carbon atoms above share the same atomic number but their mass numbers are different.

Chlorine: $^{35}_{17}Cl$, $^{37}_{17}Cl$.

All the above chlorine atoms have the same atomic number but different nucleon numbers.

Activity

Mention other elements that exhibit isotopy.

Bonding

Elements always try to achieve the stable structure or the noble gas structure and in doing so, they combine chemically forming bonds.

Types of bonds.

They are three types of bonds;

The electrovalent/ionic bonds.

Covalent bonds.

Metallic bonds.

Elements are made of atoms.

An atom is the smallest unit of an element with properties of that element.

Empirical and molecular formulae.

Empirical formula refers to a formula which shows the simplest ratio of the number of atoms of the various elements in a compound.

Molecular formula is the formula which portrays the exact number of atoms of the different elements in one molecules of a compound.

The empirical formula can be calculated from the composition by mass of the constituent elements.

Activity

1. A hydrocarbon compound contains 80% by mass of carbon and the rest being hydrogen. Calculate the empirical formula. (C=12, H=1)

Response

Elements present	**Carbon** **C**	**Hydrogen** **H**
Percentage composition	80	100-80=20
Number of moles	80/12 6.67	20/1 20
Mole ratio	6.67/6.67	20/6.67
	1	2.999 rounded off to 3
Simplest ratio	1	3

<u>The empirical formula is CH_3</u>

2. Metal N has a relative atomic mass of 197. The metal yields a chloride with 35.1% chlorine. Calculate the simplest formula of the chloride. (Cl=36)

Response

Elements present	**N**	**Chlorine** **Cl**
Percentage composition	100-35.1=64.9	35.1
Number of moles	64.9/197 0.33	35.1/36 0.98
Mole ratio	0.33/0.33	0.98/0.33
	1	2.97 rounded off to 3
Simplest ratio	1	3

<u>The empirical formula is NCl_3</u>

3. A hydrocarbon compound contains 80% by mass of carbon and the rest being hydrogen. The relative molecular mass of the compound is 30. Calculate the molecular formula of the compound. (C=12, H=1)

APPRECIATION OF THE APPLICATION OF CHEMISTRY IN DAILY LIFE

This features chapters like chemicals for consumers (food additives, drugs and medicine, detergents) and nuclear processes.

This contributes one item in section A of chemistry paper 1.

CHEMICALS FOR CONSUMERS

Food additives

Category;

Food additives can be natural or synthetic for example,

Preservatives like table salts prevent food fermentation and any microbial growth hence increasing the shelf – life of food.

Flavoring agents improve the aroma and taste of food.

Thickeners do increase the viscosity of food.

Sweeteners add sweetness to food.

Antioxidants do prevent oxidation and discoloration of oily foods.

Food dyes are used to enhance the physical appearance, textures of food.

Dangers and mitigations of food additives.

They cause allergic body reactions to some people hence skin irritations, this can be **mitigated by** using recommended amounts.

Preservatives, Stabilizers and antioxidants contain chemicals that are agents to organ cancers, hyperactivity in children, and digestive disorders especially to people who consume them in excess amounts. This can be **mitigated by** limiting their consumption rates.

Some are acid additives which lower the pH of the stomach causing health issues such as ulcers. This can be **mitigated by** proper reading on the food labels before use and regulation of consumption.

Evaluation (similarities and differences).

Similarities

Both synthetic and natural food additives are used to improve the quality of food, safety and appeal of food.

Differences

Natural food additives are less effective in action than synthetic food additives.

Natural food additives have fewer side effects than synthetic food additives.

NUCLEAR PROCESSES

Category

Nuclear processes are classified as nuclear fission and nuclear fusion.

How it works/functions.

Nuclear fission involves the splitting of an unstable nucleus into light stable nuclei by bombardment with fast moving neutrons releasing energy and more neutrons which can further propagate the reaction. The energy released can be used to generate electricity and also produce atomic bombs.

Nuclear fusion involves combination of two light nuclei to form a heavier nucleus releasing large amounts of energy. The energy generated is used to convert water to steam to run the water turbines that generate electricity, medical isotopes for diagnostics and treatment of cancer.

The energy can also make hydrogen bombs.

Calculations

The expressions below are used;

$\ln No/Nt = \lambda t$,

where;

No = initial amount,

Nt = the amount left after time, t

t = time,

λ = decay constant

For half-life;

$t_{1/2} = In2/\lambda$

The time at;

a) A/2 is t_1 where;

A is the amount of a substance and

t_1 is the time taken for the amount to decay to half the original mass.

b) A/4 is t_2 where;

A is the amount of a substance and

t_2 is the time taken for the amount to decay to a quarter its original mass.

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SECTION B

This consists of two parts that is part I and part II. Part I contains two items from the many industrial processes and part II consists of two items from natural resources.

PART I:

THE INDUSTRIAL PROCESSES

These processes portray the great contributions of chemistry to the economy of Uganda.

A student is expected to make a report/message/presentation/write up for the required industrial process with the following; raw materials, well elaborated process of production, dangers of the process of production and their mitigations, social benefits.

These include the manufacture of oxygen, biogas, ethanol, soapy and soapless detergents, cement and lime, extraction of copper, iron, aluminium, manufacture of sulphuric acid, nitric acid, fertilizers, sodium hydroxide, chlorine gas.

Manufacture of Oxygen.

There is a high demand for oxygen in national referral hospitals in Uganda. An investor was contacted by government to set up an oxygen manufacturing plant at Namanve, one of the nearby swamps to Kampala to tap into the opportunity. However, the residents seem not to understand how the process will occur, its consequences and how the project benefits them. As a knowledgeable chemistry student, you are required to create awareness to the members using the necessary information.

Task;

Write a presentation you will use upon meeting them.

Required response

Raw material; liquid air / air.

Process of production; Air is passed through air filters to remove dust and smoke particles. It is then passed through concentrated sodium hydroxide solution to remove carbon dioxide,

$2NaOH(aq) + CO_2 (g) \rightarrow Na_2CO_3(aq) + H_2O(l)$.

Air free from carbon dioxide is now passed through Silicon (IV) oxide to absorb water vapour. Carbon dioxide and water vapour are removed from air before it is liquefied because they would solidify and block the apparatus.

The air is then compressed at 200 atmospheres and allowed to cool by making it escape into a large space through a jet. The process of cooling is repeated several times to obtain liquid air at about -200^oC. The liquid air is fractionally distilled using a fractionating column / tower. Nitrogen boils off first because it has a lower boiling point (-196^oC) leaving behind oxygen with a higher boiling point (-183^oC). Pure oxygen is then stored under pressure in steel cylinders.

PART II: THE NATURAL RESOURCES

The topics listed below mostly contribute the two items in part II of section B and some concepts are examined in section A as illustrated in some of the items. They include; air, water, rocks and minerals, carbon in the environment and fossil fuels.

Natural resources are materials that occur in nature and are made use of by humans for development and survival.

They are categorized into two i.e., renewable and non-renewable natural resources.

Renewable natural resources include air, water, natural vegetation.

Non-renewable resources include rocks and fossil fuels.

AIR:

Air is a mixture of various gases.

Air is a renewable natural resource because it can be replenished or cannot be completely used up in man's lifetime.

The different gases occupy the atmosphere in varying percentages as shown below.

Gas	Chemical symbol	Percentage by volume
Nitrogen	N_2	78%
Oxygen	O_2	21%
Argon	Ar	0.9%
Carbon dioxide	CO_2	0.03 – 0.04%
Other gases (Neon, Helium etc.)	Ne, He respectively	Trace amounts

Activity:

Draw a pie chart to show the different compositions of air.

Describe the various uses of the components of air.

Air Pollution.

This occurs when excessive quantities of gases and particles are released into the atmosphere. This changes the composition of air and harms living organisms and the environment at large.

Causes or sources of air pollution and their effects or consequences.

Burning of fossil fuels. This commonly occurs in engines of vehicles, power stations and factories. A lot of gases are liberated such as carbon dioxide, carbon monoxide, sulphur dioxide, and nitrogen oxides. These cause acid rain when they mix with water, destroying walls of buildings and lowering soil pH.

Industrial processes that take place in chemical plants and refineries release various harmful particulates and gases like carbon dioxide, a greenhouse gas. This leads to respiratory illnesses like asthma, bronchitis and other lung diseases.

WATER

Water is a renewable natural resource for it can be replenished/replaced.

Occurrence of water.

Water is the most abundant substance on the earth's surface occupying approximately 71% of the planet.

It has an uneven distribution with salty water in oceans and seas taking 97.5%. This is unsuitable for use before treatment. The freshwaters take on only 2.5% of which some is in form of glaciers and ice caps, ground water, surface water and atmospheric waters.

Physical properties of water.

Water exists in all the three states of matter. It is a solid at 0^0c, a liquid at room temperature, and a gas a 100^0c.

It has a maximum density of $4gcm^{-3}$ and a high specific heat capacity i.e.; water heats up and cools down slowly.

It is colorless, odorless and tasteless.

It has a high surface tension i.e.; molecules stick together strongly.

It has a boiling point of 100^0c and a freezing point of 0^0c.

Chemical properties of water.

Water is a neutral substance with a pH of 7.

It is a universal solvent i.e.; it dissolves many substances like sugars and salts.

It reacts with metals, metal oxides and non-metals.

Importance of water.

Water is a habitat for many aquatic organisms; water bodies like lakes, rivers, swamps, dams, pools contain necessary conditions for survival of animals like snakes, worms, bacteria and plants e.g., blue green algae, planktons which are fish foods.

Water bodies like; lakes, rivers, pools, as well as water vapor from plants play a crucial role in rain formation. Water from the water bodies evaporates and eventually cools and condenses on the clouds, these result into precipitation.

Water bodies like rivers can be used to generate electricity, fast moving waters to the rivers drive turbines at waterfalls which convert kinetic energy to electrical energy.

CARBON IN THE ENVIRONMENT

A fuel is a substance that produces energy when burnt.

Carbon compounds like wood, charcoal, coal, natural gas, and oil are thus referred to as carbon-based fuels for they contain carbon and hydrogen. These burn in oxygen to give light and heat.

Wood, charcoal and all other fuels from vegetation are renewable because they can be replenished naturally within a human lifespan. The fossil fuels are not renewable because they exist in finite quantities and cannot be replaced once used.

Impacts of burning carbon-based fuels on the environment.

Air pollution caused by soot, sulphur dioxide, carbon monoxide and nitrogen oxides. This leads to smog, acid rain, and respiratory illnesses like asthma, bronchitis.

Greenhouse gas emissions due to increased concentrations of carbon dioxide, methane hence global warming and climate change.

Ocean acidification due to absorption of carbon dioxide by oceans to form carbonic acid. This harms the coral reefs and shellfish.

Deforestation due to unsustainable harvesting of firewood and wood for charcoal. This leads to soil erosion and loss of biodiversity.

Acid rain due to reaction of water with sulphur and nitrogen oxides in the atmosphere. This damages forests, soils and buildings.

Worked out Item.

Charcoal is a common fuel used in many homes in Uganda. During its combustion in excess oxygen supply, it produces carbon dioxide. Your friend is using 80.5g of charcoal to cook food in a busy evening for guests. It burns completely as shown in the equation below.

$C(s) + O_2 (g) \longrightarrow CO_2 (g)$

Task;

As a learner of chemistry

a) Assess the product formed.

The product is carbon dioxide, a covalent compound, because it is formed by sharing of electron between non-metal atoms, examples include sulphur dioxide, carbon dioxide.

b) Advise the friend on the appropriateness of the product formed.

Carbon dioxide has the following **properties;**

It sparingly dissolves in water; has a slightly sour taste; it is colourless, odourless and denser than air.

The product **is used in**; - fire extinguishers to put out fire; the manufacture of fizzy drinks to improve on the taste and to preserve them.

Combustion of oil, releases carbon dioxide which can result into acid rain that interferes with the soil pH, hence affecting plant growth, it also leads to global warming and climate change. This is **mitigated by** increased afforestation such that the trees absorb carbon dioxide.

Oil spills by humans can reach water bodies cutting off oxygen supply and causing suffocation of aquatic animals. This is **mitigated by** proper waste management routines.

Importance of the natural resource.

When purified it produces various chemicals like motor fuels, lubricants, paints, detergents, drugs, etc.

Fossil fuels are used as fuels, when burnt they produce heat energy used to run engines and machines, even for cooking.

Sample Item.

The setup of crude oil mines in Hoima City resulted in displacing of residents to other places but the victims are still not contented. This has made the government to intervene. The government is planning to sensitize the residents through workshops in the city.

Task:

As a chemistry student prepare a short presentation you will deliver during the workshop upon invitation.

END